

WORK-IN-PROGRESS REPORT

Product Sentiment Analysis and Feature Roadmap Prioritization Using Python

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1. Updated Problem Statement & Scope

The project focuses on analyzing user reviews of Health & Fitness applications to understand customer sentiment, identify product concerns, and support data-driven feature prioritization. Based on the initial implementation, the scope has been refined into a structured pipeline covering app filtering, review cleaning, sentiment analysis, feature extraction, and RICE-based prioritization.

The refined scope includes cleaning Google Play Store app and review data, filtering **Health & Fitness** applications, selecting the **top 20–30 apps based on review count with a minimum threshold of 100 reviews**, preprocessing user reviews, performing VADER sentiment analysis, extracting feature requests using NLP/TF-IDF, prioritizing features using the RICE framework, and presenting insights through visualizations and a dashboard.

2. Implementation Status

Module	Status	Progress
Data Import & Inspection	✓ Completed	Loaded and inspected both datasets
Review Data Cleaning	✓ Completed	Removed missing, short and duplicate reviews
App Metadata Cleaning	✓ Completed	Cleaned key fields and duplicate app records
Health & Fitness Filtering	✓ Completed	Identified 254 Health & Fitness apps
Top App Selection	✓ Completed	Selected top 30 apps based on review count
Dataset Matching	🔄 In Progress	Validating app-name matching before merging
VADER Sentiment Analysis	⌚ Pending	To be implemented after final dataset validation
Feature Extraction	⌚ Pending	NLP/TF-IDF analysis
RICE Prioritization	⌚ Pending	Feature ranking
Dashboard	⌚ Pending	Visualization and recommendations

```
print("FINAL CLEANING SUMMARY")
print("=" * 40)

print("Original records :", len(reviews))
print("Clean records    :", len(reviews_clean))
print("Records removed  :", len(reviews) - len(reviews_clean))
print("Unique apps       :", reviews_clean['App'].nunique())

print("\nMissing values:")
print(reviews_clean.isnull().sum())

... FINAL CLEANING SUMMARY
=====
Original records : 64295
Clean records    : 29692
Records removed  : 34603
Unique apps      : 865

Missing values:
App              0
Translated_Review 0
Sentiment        0
Sentiment_Polarity 0
Sentiment_Subjectivity 0
dtype: int64
```

Figure 1. Review Data Cleaning — Code & Output

```
health_fitness = apps_clean[
    apps_clean['Category'] == 'HEALTH_AND_FITNESS'
].copy()

print("Health & Fitness apps:", len(health_fitness))

Health & Fitness apps: 288

health_fitness = health_fitness[
    health_fitness['Reviews'] >= 100
].copy()

print("Health & Fitness apps with 100+ reviews:", len(health_fitness))

Health & Fitness apps with 100+ reviews: 201

top_apps = health_fitness.sort_values(
    by='Reviews',
    ascending=False
).head(30).copy()
```

Figure 2. Health & Fitness Filtering & Top 30 Selection — Code & Output

3. Partial Output / Test Results

Test / Output	Result
Original review records	64,295
Clean reviews after preprocessing	27,307
Health & Fitness applications	254
Applications selected for analysis	30
Exact app-review matches	8

Table 1. Partial Output Summary

Partial Output:

The preprocessing and filtering pipeline is working successfully. The review dataset was reduced to **27,307 usable**

reviews, and **30 Health & Fitness applications** were selected based on review count. Initial dataset matching produced **8 exact app-name matches**; the remaining mismatches are being investigated before the final merge and sentiment analysis.

Working Test Case:

The current pipeline successfully performs:

Raw Reviews → Cleaning → Health & Fitness Filtering → Top-30 Selection → App Matching

```
# Check which selected apps have matching reviews
matched_apps = set(top_apps['App']).intersection(
    set(reviews_clean['App'])
)

print("Top 30 apps:", len(top_apps))
print("Apps with matching reviews:", len(matched_apps))
print("Apps without matching reviews:", len(top_apps) - len(matched_apps))

... Top 30 apps: 30
Apps with matching reviews: 8
Apps without matching reviews: 22

# Display apps that have matching reviews
matching_app_list = sorted(matched_apps)

for app in matching_app_list:
    print(app)

30 Day Fitness Challenge - Workout at Home
7 Minute Workout
Calorie Counter - MyFitnessPal
Fabulous: Motivate Me! Meditate, Relax, Sleep
Fitbit
Garmin Connect™
Google Fit - Fitness Tracking
Home Workout - No Equipment
```

Figure 3: Partial Output / Test Result

4. Challenges Faced

Challenge	Resolution / Current Action
Incorrect metadata file	Replaced the incorrectly loaded selected-columns file with the original Google Play Store dataset.
Missing & short reviews	Removed null/empty reviews and reviews below the 10-character threshold during preprocessing.
Duplicate records	Removed duplicate reviews and duplicate app records to improve data quality.
App-name mismatch	Exact matching currently identifies 8 of 30 selected apps; name standardization is being investigated before final merging.

Table 2. Challenges Faced

5. Remaining Work Plan

Task	Target Date
Finalize app matching & dataset merge	23 Aug
VADER analysis & testing	25 Aug
NLP/TF-IDF feature extraction	28 Aug
RICE prioritization & dashboard	31 Aug
Final testing, report & presentation	5 Sep

Table 3. Remaining Work Plan

6. Summary

The project has successfully completed the initial **data preparation and filtering stages**. Both Google Play Store datasets were loaded and reviewed, **64,295 reviews were cleaned to 27,307 usable records**, and **254 Health & Fitness applications** meeting the minimum review requirement were identified. The **Top 30 applications** were selected for further analysis. Initial app-review matching has been performed, with remaining name inconsistencies being addressed.

The next phase will focus on **VADER sentiment analysis, NLP/TF-IDF feature extraction, RICE prioritization, dashboard development, and final testing** before the final submission on **5 September 2026**.